Safe Sharing of Genomic Data from Discovery to Broad Data Sharing

Leveraging the Power of Beacon and Privacy Enhancing Technology (PET)

Soyean Kim*[1], Koohyung Han [3], Mahdi Mobini [4], Gordon Ritchie [4], Dan Holmes [4] , Jun Duan [1], Jihoon Cho [3], Hyojin Yoon [3], Bjorn Gruning [5]; Caroline Colijn [4], William Hsiao[1,4]

[1]Faculty of Health Sciences, Simon Fraser University; [2]Department of Mathematics, Simon Fraser University ; [4]Department of Molecular Biology and Biochemistry, Simon Fraser University; [3] Samsung SDS; [4]St. Paul's pathology lab; [5]University of Freiburg; *corresponding author

INTRODUCTION

- While pathogen genomic data is generally considered to be non-identifying, the associated contextual data and the potential of amplifying the human host genetic sequences are potentially sensitive. It is often the cumulation and combination of indirect identifiers that makes a specific dataset identifiable. While data sharing of pathogen genomic data in a timely manner has a direct impact on our collective health, real-time sharing of such data is a challenge due to discoverability and privacy issues
- Galaxy has gained a Beacon integration that enables users to publish genetic variance data conveniently via Beacon project. The Beacon network enables querying genomic variant catalogues without the risk of exposing sensitive data Beacon facilitates the first-contact and speeds up knowledge exchange dramatically.
- In the infectious diseases context using microbial genomics data, lack of the reference genomes has been a challenge to allow mutation-based data discovery until recently. The latest development in the pathogen genomics data ecosystem, thanks to the COVID-19 pandemic, offers a promising roadmap on how pathogen genomics data discovery can be enhanced using tools such as Beacon based on a mutation.
- Privacy Enhanced Technologies (PETs) are emerging technologies that can provide new ways for data sharing. Secure Multiparty Computation (SMC) is one of the PETs, and supports computations of inputs from different organizations without revealing each party's input. Private Set Intersection (PSI) and Linear secret sharing (LSS) are commonly used methods in SMC

METHODS

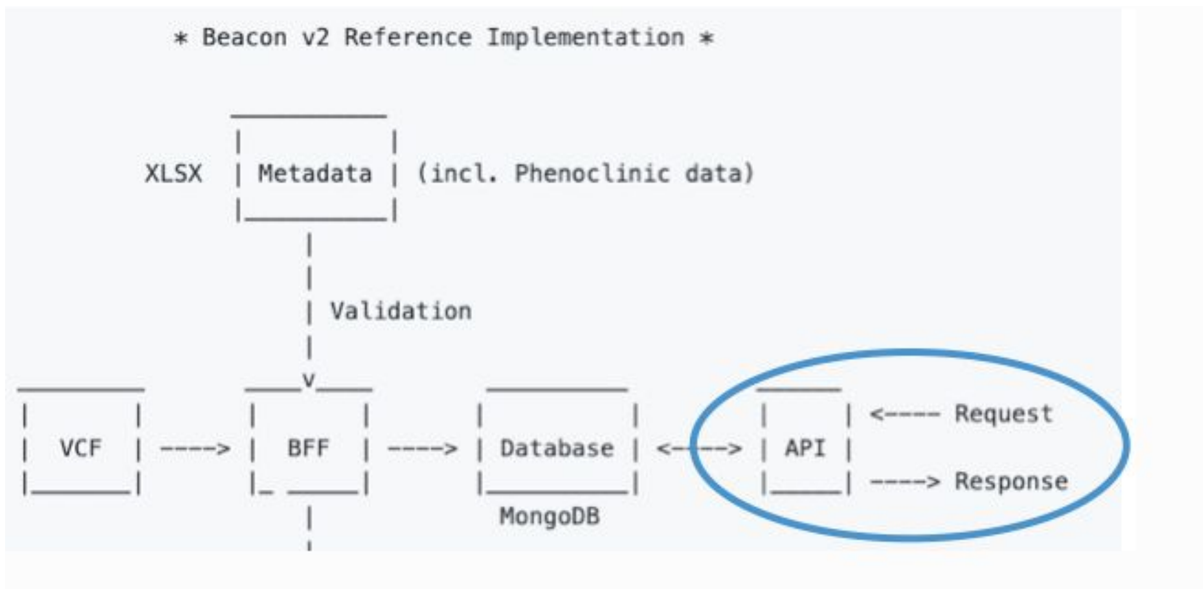

Figure 3. Overview of the data ingest and validation steps

METHODS

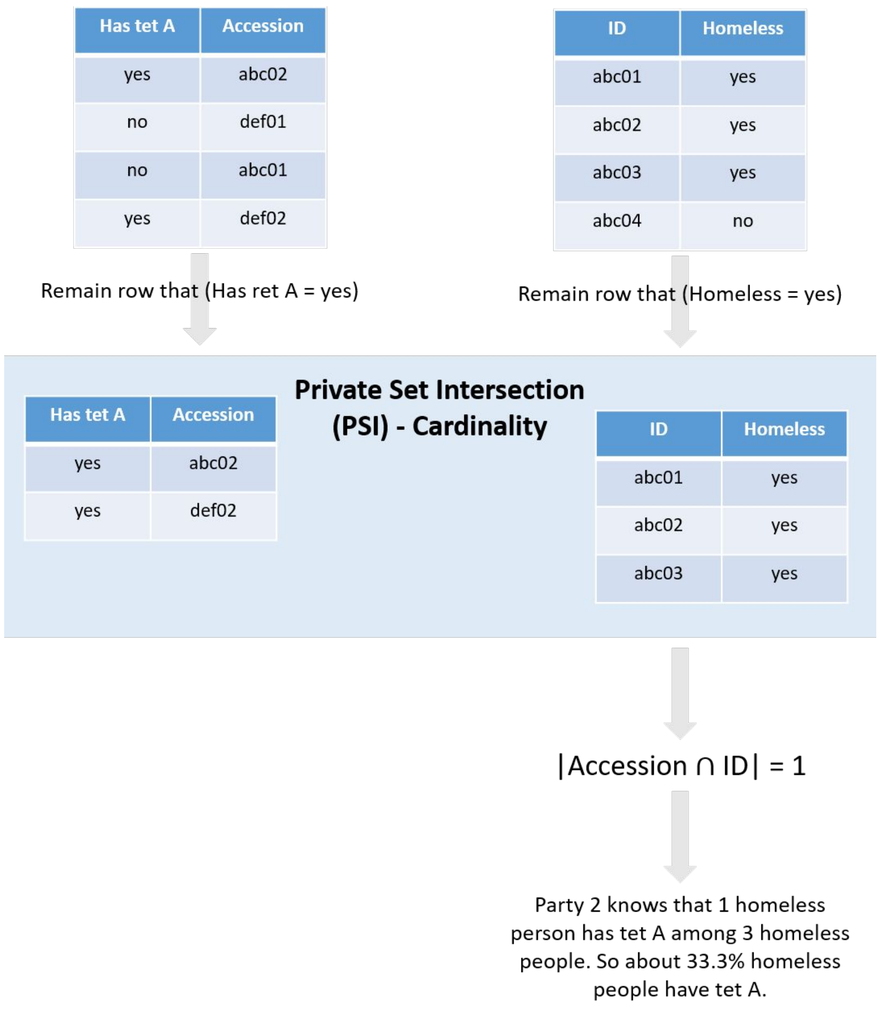

Figure 1. PSI data linking step is shown post the first contact initiated by Beacon in Galaxy. A researcher may want to ask the network if something about relationship between the tetracycline resistance gene and homeless people. The tet (A) gene is present in Shigella bacteria that is causing infections in the community. The researcher gets a percentage of homeless people with tet(A) gene which is 33.3% in the figure above. Similarly, broader metadata such as travel status can be linked using PSI.

RESULTS

Figure 4. screenshots for Beacon deployment showing data entry, search by phenotype and cohort data annotated with ontologies

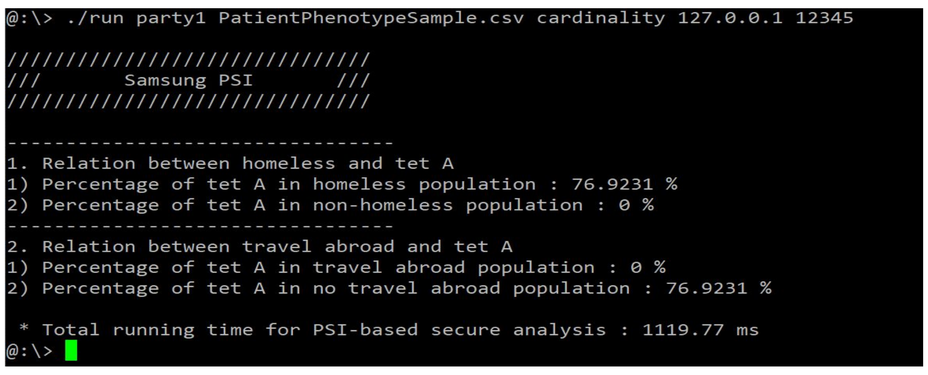

Figure 5. screenshots for PSI based secure analysis. As a result, we figure out that 10 homeless people has Tet A gene among 13 (which is 76.92% vs. 0% in non-homeless population). This experiment is done at local network in Linux PC.Total running time for PSI-based secure analysis : 1077.47 ms

Impact of broader data sharing in the analysis output

- Pathogen genome only clustering is too broad – additional metadata is needed for better clustering and targeting intervention (i.e. drug susceptibility patterns in different population groups)
- Valuable and sensitive data in the hospitals can be linked such as homeless status, travel history, the reason for testing, serology, and vaccination status.

ACKNOWLEDGEMENTS

canarie, GenomeCanada, Genome BritishColumbia

ska8@sfu.ca, kh89.han@Samsung.com

CINECA — Common Infrastructure for National Cohorts in Europe, Canada, and Africa

CONCLUSIONS

- In this study, we explored the feasibility of broader data sharing of pathogen genomics by combining Beacon and Privacy-enhancing technology.
- Together with the Beacon project, SMC could expand the scope of analysis without privacy issue. Combination of personal information (e.g. housing status) with pathogen genomic data is useful, but the data owners (especially in health care) may be hesitant to share their data. SMC can be used for collaborative analysis after the first contact made by the Beacon project. Broader sharing of contextual data holds promise to address critical public health issues such as understanding distinct epidemiological profiles emerging from sub populations.